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Chapter #10

1. What is a GUI?
 - a. A GUI (Graphical User Interface) is a way for a user to interact with an application using buttons, icons, images, and other visual elements instead of using text commands
2. Explain how code is executed in an event-driven application
 - a. Rather than the program just running in a set order/sequence, it runs as a loop and only acts when it gets an external prompt from the user via a button or other possible actions.
3. Can components be added directly to a frame? Explain.
 - a. No, components are added to the content Pane
4. Can a label respond to events? Explain.
 - a. Yes, although they don't typically have an action listener, they can be configured to detect mouse interactions
5. Why do you think a GUI needs to be run from an event-dispatching thread?
 - a. Yes, it keeps things from running at the same time and interfering with each other.
6. What is the difference between a label and a button?
 - a. A button is designed to be interacted with and respond to user input, while a Label is used to output visuals
8. List three ways to control the layout of a control pane.
 - a. Absolute positioning
 - b. Layout managers
 - c. Nested panels
13. What must first be done with numeric data typed in a text field before it can be used in calculation?
 - a. It must be converted from a string.
14. What is the value of num1 in the last statement below?

```
double num1;  
Double num2 = new Double(3);  
String num3 = "5";  
num1 = num2.doubleValue() + Double.valueOf(num3).doubleValue();
```

 - a. num1 is 8.0

15. An application prompts a user to select a name from a list of six names. Which is a better component choice: a text field or a combo box? Explain.
- a. A Combo Box is a better choice because it prevents the user from inputting typos because they're limited to the option in the combo box